

Midterm #1

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UNIT 0

1. c. 11.0 g cm^{-3}

2. $t = d/v$

$d = 600 \text{ km}$

$v = 60 \text{ km/hr}$

$t = \frac{600 \text{ km}}{60 \text{ km/hr}} = 10 \text{ hrs.}$
c.

3. $\frac{25 \text{ m}}{\text{s}} \times \frac{3600 \text{ s}}{1 \text{ hr}} \times \frac{1 \text{ km}}{1000 \text{ m}} = 90 \text{ km/hr}$
d.

4. $v_i = 0 \text{ km/hr}$

$v_f = 10 \text{ km/hr}$

$\Delta t = 60 \text{ s} = \frac{1}{60} \text{ hr}$

$a = \frac{\Delta v}{\Delta t} = \frac{10 \text{ km/hr}}{\frac{1}{60} \text{ hr}} = 600 \text{ km/hr}^2$

$\downarrow$
 $\frac{600 \text{ km}}{\text{hr}^2} \div 3600 \text{ s} = \frac{600}{3600} \text{ km/hr/s} = \frac{1}{6} \text{ km/hr/s}$
c.

5. b. 5000 cm^2

6. $1 \text{ L} = 1000 \text{ cm}^3$, $2 \text{ L} = 2000 \text{ cm}^3$ coffee bean: 0.5 cm^3

$\frac{\text{volume of bottle}}{\text{volume of coffee bean}} = \frac{2000 \text{ cm}^3}{0.5 \text{ cm}^3} = 4000 \text{ or } 4 \times 10^3$ c.

7. $\vec{v} = v_x \hat{i} + v_y \hat{j}$

$v_x = v \cos(\theta)$

$v_y = v \sin(\theta)$

$v = 10 \text{ km/hr}$

$\theta = 45^\circ$

$\cos(45^\circ) = \sin(45^\circ) = 1/\sqrt{2} = 0.7071$

$v_x = 10 \text{ km/hr} \times 0.7071 = 7.071 \text{ km/hr}$

$v_y = 10 \text{ km/hr} \times 0.7071 = 7.071 \text{ km/hr}$

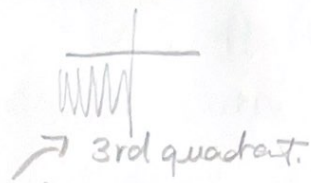


negative quadrant = $(-7.1 \text{ km/hr}, -7.1 \text{ km/hr})$
d.

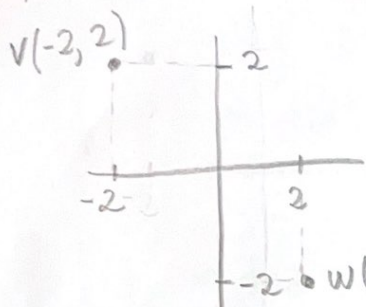
8. $v_x = -7.1 \text{ km/hr}$ vector is in the
 $v_y = -7.1 \text{ km/hr}$, third quadrant

$$\tan(\theta') = \frac{|v_y|}{|v_x|} = 1 \quad (\theta') = 45^\circ, \text{ or } 180^\circ$$

$$\theta = 180^\circ + \theta' = 180^\circ + 45^\circ = 225^\circ$$



9. coordinate system visualization:



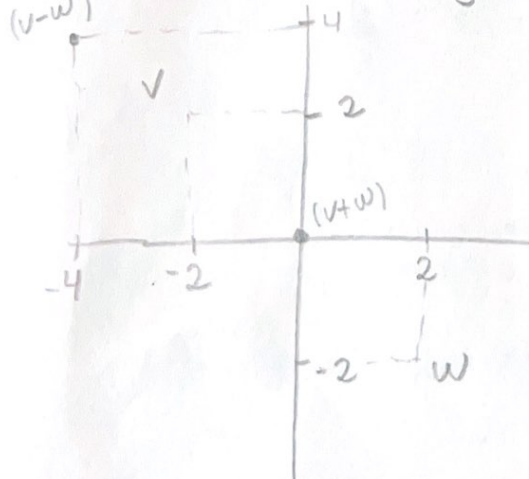
$$\begin{aligned} \vec{v} &= -2\hat{i} + 2\hat{j} \\ \vec{w} &= 2\hat{i} - 2\hat{j} \end{aligned}$$

$$\vec{v} + \vec{w} = 0\hat{i} + 0\hat{j} \quad (0, 0)$$

$$\begin{aligned} \vec{v} &= -2\hat{i} + 2\hat{j} \\ -\vec{w} &= 2\hat{i} - 2\hat{j} \end{aligned}$$

$$\vec{v} - \vec{w} = -4\hat{i} + 4\hat{j} \quad (-4, 4)$$

new coordinate system:



$$\vec{v} \times \vec{w} = v_x w_y - v_y w_x$$

$$\vec{v} \times \vec{w} = (-2)(-2) - (2)(2)$$

$$\vec{v} \times \vec{w} = 4 - 4$$

$$\vec{v} \times \vec{w} = 0$$

UNIT 1

1. $v_0 = 15 \text{ m/s}$
 $a = 3 \text{ m/s}^2$
 $t = 4 \text{ s}$

$$v = v_0 + a \cdot t$$
$$v = 15 \text{ m/s} + (3 \text{ m/s}^2) \cdot 4 \text{ s}$$
$$v = 15 \text{ m/s} + 12 \text{ m/s} = 27 \text{ m/s}$$

$$S = v_0 \cdot t + \frac{1}{2} a t^2$$

$$S = 15 \text{ m/s} \cdot 4 \text{ s} + \frac{1}{2} 3 \text{ m/s}^2 \cdot 4 \text{ s}^2$$

$$S = 60 \text{ m} + \frac{1}{2} \cdot 3 \cdot 16$$

$$S = 60 \text{ m} + 24 \text{ m}$$

$$S = 84 \text{ m}$$

@ $t = 0$

$$\text{average velocity} = \frac{v_0 + v}{2} = \frac{15 \text{ m/s} + 15 \text{ m/s}}{2} = 15 \frac{\text{m}}{\text{s}}$$

@ $t = 4 \text{ s}$

$$\text{average velocity} = \frac{\text{displacement}}{\text{time}} = \frac{84 \text{ m}}{4 \text{ s}} = 21 \text{ m/s}$$

2. P $\rightarrow t = 15 \text{ s}, x = 988 \text{ m}$ Q $\rightarrow t = 20 \text{ s}, x = 1500 \text{ m}$

Speed @ P

$$x(t=10) = 600 \text{ m}, x(t=15) = 988 \text{ m}$$

$$v_p = \frac{x(15) - x(10)}{15 \text{ s} - 10 \text{ s}} = \frac{988 \text{ m} - 600 \text{ m}}{5 \text{ s}} = \frac{388 \text{ m}}{5 \text{ s}} = 77.6 \text{ m/s}$$

speed @ Q

$$v_q = \frac{x(20) - x(15)}{20 \text{ s} - 10 \text{ s}} = \frac{1500 \text{ m} - 988 \text{ m}}{5 \text{ s}} = \frac{512 \text{ m}}{5 \text{ s}} = 102.4 \text{ m/s}$$

acceleration @ P & Q:

$$\Delta v = v_q - v_p = 102.4 \text{ m/s} - 77.6 \text{ m/s} = 24.8 \text{ m/s}$$

$$\Delta t = 20 \text{ s} - 15 \text{ s} = 5 \text{ s}$$

$$a = \frac{\Delta v}{\Delta t} = \frac{24.8}{5} = 4.96 \text{ m/s}^2$$

3. $u = 0 \text{ m/s}$

$v = 6 \text{ m/s}$

$a = 0.8 \text{ m/s}^2$

$$S = \frac{(6)^2 - (0)^2}{2 \times 0.8}$$

$$S = \frac{36}{1.6} = 22.5 \text{ m}$$

$$v^2 = u^2 + 2as$$

$$S = \frac{v^2 - u^2}{2a}$$

$$t = \frac{6 - 0}{0.8}$$

$$t = \frac{6}{0.8} = 7.5 \text{ s}$$

$$v = u + at$$

$$t = \frac{v - u}{a}$$

4. $R = \frac{v_0^2 \sin(2\theta)}{g}$

$R = 60 \text{ m}$, $g = a = 9.81 \text{ m/s}^2$

1. @ 45°

$$R = \frac{v_0^2 \sin(90^\circ)}{9.81 \text{ m/s}^2}$$

$$60 \text{ m} (9.81 \text{ m/s}^2) = v_0^2 \sin(90)$$

$$v_0^2 = 588.6 = 24.25 \text{ m/s}$$

for 45° :

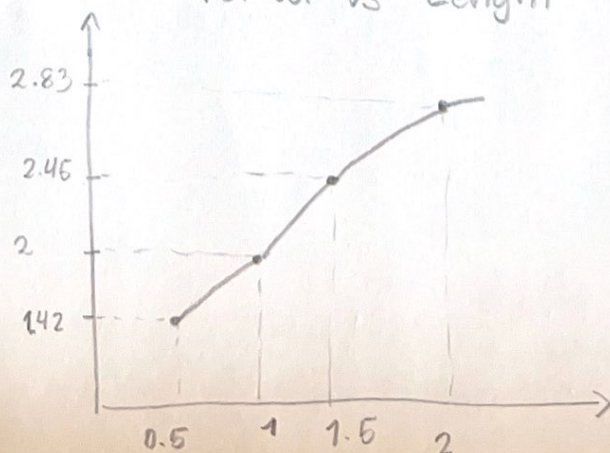
$$T = \frac{2v_0 \sin(\theta)}{9.81 \text{ m/s}^2}$$

$$T = \frac{2 \times 24 \times \sin(45)}{9.81} = \frac{34.29}{9.81} = 3.49 \text{ s}$$

5.

L (m)	Period T (s)	T ² (s ²)	g (m/s ²)
0.5 m	1.42 s	2.0164 s ²	g ₁ = 9.81
1 m	2 s	4 s ²	g ₂ = 9.87
1.5 m	2.45 s	6.0025 s ²	g ₃ = 9.80
2 m	2.83 s	8.0089 s ²	g ₄ = 9.82

Period vs Length



Work in pairs to measure the period of a pendulum of different lengths and compare your results with the theoretical values.

UNIT 2

$$1. F_{\perp} = 1000 \text{ N} \quad F_{\perp} = T \cdot \sin \theta \quad T = \frac{F_{\perp}}{\sin \theta} = \frac{1000}{\sin(7^{\circ})} = 8192.6 \text{ N}$$
$$\theta = 7^{\circ}$$

$$m = 900 \text{ kg} \quad N = W = m \cdot g$$
$$\mu_k = 0.05 \quad W = 900 \text{ kg} \times 9.81 \text{ m/s}^2 = 8829 \text{ N}$$

$$F_f = \mu_k \cdot N = 0.05 \times 8829 \text{ N} = 441.45 \text{ N}$$

$$F_{\text{net}} = T - F_f = 8192.57 \text{ N} - 441.45 \text{ N} = 7751.12 \text{ N}$$

$$a = \frac{F_{\text{net}}}{m} = \frac{7751.12 \text{ N}}{900 \text{ kg}} = 8.61 \text{ m/s}^2$$

$$2. m = 20,000 \text{ kg}$$
$$v_i = 120 \text{ km/hr} (33.33 \text{ m/s})$$
$$v_f = 0 \text{ m/s}$$
$$d = 100 \text{ m}$$

$$v_f^2 = v_i^2 + 2ad$$
$$a = \frac{v_f^2 - v_i^2}{2d}$$

$$a = \frac{0^2 - (33.33 \text{ m/s})^2}{2 \cdot 100 \text{ m}} = \frac{-1111.09 \text{ m}^2/\text{s}^2}{200 \text{ m}} = -5.56 \text{ m/s}^2$$

$$b) F = m \cdot a$$

$$F = 20,000 \text{ kg} \times -5.56 \text{ m/s}^2 = -111,200 \text{ N}$$

$$3. F_1 = 10 \text{ N} @ 45^{\circ}$$

$$F_1:$$

$$F_2 =$$

$$F_2 = 8 \text{ N} @ 30^{\circ}$$

$$F_{1x} = F_1 \cos 45^{\circ}$$

$$F_{2x} = F_2 \cos 30^{\circ} = 6.93 \text{ N}$$

$$F = 7.5 \text{ N}$$

$$10 (\sqrt{2}/2) = 7.07 \text{ N}$$

$$m = 50 \text{ kg}$$

$$F_{1y} = F_1 \sin 45^{\circ} = 7.07 \text{ N}$$
$$10 (\sqrt{2}/2)$$

$$F_{2y} = F_2 \sin 30^{\circ} = 4 \text{ N}$$

$$F_x = F = 7.5 \text{ N}$$

$$F_{\text{net}x} = F_{1x} + F_{2x} + F_{3x}$$

$$F_{\text{net}y} = F_{1y} + F_{2y}$$

$$F_y = 0 \text{ N}$$

$$F_{\text{net}x} = 7.07 \text{ N} + 6.93 \text{ N} + 7.5 \text{ N}$$

$$F_{\text{net}y} = 7.07 + 4 \text{ N}$$

$$F_{\text{net}x} = 21.5 \text{ N}$$

$$F_{\text{net}y} = 11.07 \text{ N}$$

$$a = \frac{F_{\text{net}}}{m} = \frac{21.50 \text{ N}}{50 \text{ kg}} = 0.43 \text{ m/s}^2$$

UNIT 3

1. Deriving from acceleration $a = g(\sin\theta - \mu\cos\theta)$

$$W = mg, \quad N = \text{normal force}, \quad F = \mu N$$

$$W_{\parallel} = mg \sin\theta$$

$$W_{\text{perpendicular}} = mg \cos\theta$$

$$N = W_{\perp} = mg \cos\theta$$

$$F = \mu N = \mu (mg \cos\theta)$$

$$F_{\text{net}} = mg \sin\theta - \mu mg \cos\theta$$

$$a = g(\sin\theta - \mu \cos\theta)$$

$$\rightarrow \text{as } \mu \rightarrow 0$$

$$a = g \sin\theta$$

2. $\theta = 10^\circ$

$$g = 9.81 \text{ m/s}^2$$

(a) $\mu = 0.1$

$$a = g(\sin\theta - \mu \cos\theta)$$

$$a = 9.81 \text{ m/s}^2 (\sin(10) - 0.1 \cos(10))$$

$$a = 9.81 \text{ m/s}^2 \times 0.07512$$

$$a = 0.738 \text{ m/s}^2$$

(b) $d = v_0 t + \frac{1}{2} a t^2$

$$d = 0 + \frac{1}{2} (0.738) (30^2)$$

$$d = \frac{1}{2} (0.738) (900) = 331.5 \text{ m}$$

$$v = v_0 + at$$

$$v = 0 + (0.738 \text{ m/s}^2) (30 \text{ s})$$

$$v = 22.14 \text{ m/s}$$

3. $m = 6000 \text{ kg}$

$$\theta = 30^\circ$$

$$L = 80,000 \text{ N}$$

$$v = 600 \text{ km/hr} (166.67 \text{ m/s})$$

(a) $W = mg = 6000 \text{ kg} \times 9.81 \text{ m/s}^2 = 58,860 \text{ N}$

$$F_c = F_{\text{net}} = L - W = 80,000 \text{ N} - 58,860 \text{ N} = 21,140 \text{ N}$$

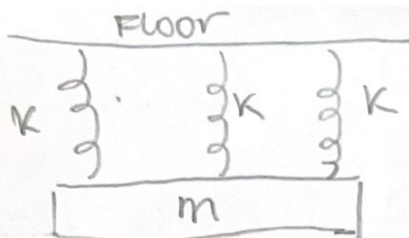
(b) $r = \frac{6000 \text{ kg} \cdot (166.67 \text{ m/s})^2}{21,140 \text{ N}} = 7875.6 \text{ m}$

(c) $d = \frac{1}{2} \cdot 2\pi r$

$$d = \pi \cdot 7875.6 \text{ m} = 24,736.73 \text{ m}$$

$$t = \frac{d}{v} = \frac{24,736.73 \text{ m}}{166.67 \text{ m/s}} = 148.8 \text{ s}$$

4. a)



b, $F_{\text{total}} = mg = 3k\Delta x$

$$\Delta x = \frac{mg}{3k}$$

c. " $k \rightarrow \infty$ " $\rightarrow \lim_{k \rightarrow \infty} \Delta x = \lim_{k \rightarrow \infty} \frac{mg}{3k}$

$$\lim_{k \rightarrow \infty} \Delta x = 0$$

5. $m = 60 \text{ kg}$

a. $g = 9.81 \text{ m/s}^2$

$\rho = 1.2 \text{ kg/m}^3$

$c = 0.5$

$A = 0.25 \text{ m}^2$

$$v_t = \sqrt{\frac{2mg}{\rho c A}}$$

$$v_t = \sqrt{\frac{2(60 \text{ kg})(9.81 \text{ m/s}^2)}{(1.2 \text{ kg/m}^3)(0.5)(0.25 \text{ m}^2)}} = \sqrt{\frac{1176}{0.15}} \frac{\text{m}}{\text{s}}$$

$$v_t = 88.52 \text{ m/s}$$

b. $A' = 100 \times 0.25 = 25 \text{ m}^2$

$$v'_t = \sqrt{\frac{2mg}{\rho c A'}}$$

$$v'_t = \sqrt{\frac{1176 \text{ kg m/s}^2}{(1.2 \text{ kg/m}^3)(0.5)(25 \text{ m}^2)}} = \sqrt{\frac{1176}{15}} \frac{\text{m}}{\text{s}}$$

$$v'_t = 8.85 \text{ m/s}$$

6. a) $d = 20 \text{ cm}$
 $F = 10,000 \text{ N}$

$L = 10 \text{ m}$

$A = \pi r^2$

$E = 45 \times 10^9 \text{ N/m}^2$

$r = 20 \text{ cm} / 2 = 10 \text{ cm} = 0.1 \text{ m}$

$A = \pi r^2 = \pi (0.1)^2 \text{ m} = 0.0314 \text{ m}^2$

$$\Delta L = \frac{FL}{AE} = \frac{10,000 \text{ N} \times 10 \text{ m}}{0.0314 \text{ m}^2 \times 45 \times 10^9 \text{ N/m}^2} = 7.08 \times 10^{-5} \text{ m}$$

b) $E_{\text{new}} = \frac{E}{2} = 22.5 \times 10^9 \text{ N/m}^2$

$\Delta L_{\text{new}} = \frac{FL}{AE_{\text{new}}}$

$$\Delta L_{\text{new}} = \left(\frac{10,000 \text{ N} \times 10 \text{ m}}{(0.0314 \text{ m}^2)(22.5 \times 10^9 \text{ N/m}^2)} \right)$$

$$\Delta L_{\text{new}} = 1.41 \times 10^{-4} \text{ m}$$